



QA Report

Big Bad Wolf Saloon — Wild-West Video Slot



Document	QA Report
Project	Big Bad Wolf Saloon
Type	Internal documentation
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TITLE	BIG BAD WOLF SALOON
Build under test	Production build deployed to super-lolly-c99fd8.netlify.app (commit 691d4f1 line)
Test date	June 1, 2026
Prepared by	QA pass (automated functional + headless browser + Monte-Carlo math verification)
Game type	Free, virtual-credit casino-style slot — no real-money wagering
Result	PASS — no blocking or major defects found; advisory items noted

1. Executive Summary

This report documents a structured Quality Assurance pass over Big Bad Wolf Saloon, covering game mathematics, core gameplay, the bonus feature, audio, user interface, the help/rules screens, interactive content, developer tools, compatibility, performance, accessibility and responsible-play compliance. Tests were executed against the current production build using a headless browser driving the live game, a Node-based Monte-Carlo simulator for the mathematics, and static analysis of the source for the items that cannot be observed at runtime.

Overall verdict: PASS. Every core gameplay and feature path completed correctly, the return-to-player matched the configured targets within normal sampling tolerance, and no JavaScript errors were observed across the entire test battery. A small number of low-severity advisory observations and recommendations (chiefly: validation on a physical iOS Safari device) are listed in Section 12.

Results at a glance

TEST / CHECK	RESULT	EVIDENCE & NOTES
GAME MATHEMATICS (RTP / HIT RATE / TRIGGER)	PASS	STANDARD 97.07% AND LEAN 84.59% OVER 10,000,000 SIMULATED SPINS EACH — WITHIN ~0.4% OF THE 97% / 85% TARGETS.
Core gameplay (spin, bet, balance)	PASS	Bet deduction, win crediting, balance maths and bet clamping all exact.
Bonus feature (incl. Buy Bonus)	PASS	Full free-spins round executed end-to-end; award credited correctly.
Audio (SFX / music / voice)	PASS	All clips present and wired; volume controls and mute/reset verified.

TEST / CHECK	RESULT	EVIDENCE & NOTES
User interface & drawers	PASS	Public + developer drawers render correctly with no overlap or overflow.
Help / Rules / Paytable	PASS	All content traceable to the par sheet; values match the math.
Interactive content / easter eggs	PASS	Clickable hotspots, symbol jokes and the High-Noon sequence all fire.
Developer tools	PASS	All 12 tools open / act; gated behind ?dev=1.
SEO & metadata	PASS	Complete head metadata, structured data, sitemap and robots.
Performance / load	PASS	First-play download ~16.7 MB; the bulk of media is deferred.
Accessibility	PASS	Crawlable text, ARIA labels and image alt text present.
Responsible-play compliance	PASS	Framed as free / virtual; no misleading or pressuring language.
Stability (runtime errors)	PASS	Zero console errors across all executed tests.
Physical iOS Safari device	INFO	Recommended — see Section 12. Browser tests used an emulated iPhone-landscape viewport, not real WebKit hardware.

2. Test Environment & Methodology

FUNCTIONAL / UI	HEADLESS CHROMIUM DRIVING THE LIVE GAME SERVED LOCALLY (TOOLS/SERVE.JS), VIEWPORT 932×430 CSS PX EMULATING AN IPHONE 16 PRO MAX HELD IN LANDSCAPE.
Mathematics	Node.js Monte-Carlo verifier (tools/sim.js) running the exact game math module (src/math/mathcore.js) — the same code the browser uses.
Static analysis	Source review of the ES-module sources in src/, the build (tools/build.js), index.html and styles.css.
Build	node tools/build.js bundles 37 ES modules (6,432 lines) to app.js. Clean build, no warnings.

Test result legend:

- PASS** — behaved exactly as expected.
- WARN** — works, but with a minor caveat worth a human look.

- **INFO** — advisory / recommended follow-up, not a defect.
- **FAIL** — defect (none found in this pass).

3. Game Mathematics & Fairness

The mathematics were verified by simulating ten million spins per model with the production math code, then comparing the measured return-to-player, hit frequency, bonus-trigger rate and bonus-buy fairness against the configured targets and the values shown to the player.

Standard model (target 97%) — 10,000,000 spins

TOTAL RTP	97.07% (TARGET 97% — WITHIN TOLERANCE)
Base game / Bonus split	49.32% base + 47.74% bonus
Hit frequency	26.42% of spins win
Bonus trigger rate	1 in 179 spins
Average free spins / bonus	7.3
Mansion jackpots	0.235 per bonus
Volatility (σ , per-spin $\times$ bet)	9.27 (high volatility)
Average win	0.9707 $\times$ bet
Largest win observed	1,239 $\times$ bet
Bonus-buy fair price	88.12 $\times$ bet (configured price 88 $\times$ — effectively fair, ~97% buy RTP)

Lean model (target 85%) — 10,000,000 spins

TOTAL RTP	84.59% (TARGET 85% — WITHIN TOLERANCE)
Hit frequency	26.42% (identical to Standard)
Bonus trigger rate	1 in 181 (identical reels)
Largest win observed	871 $\times$ bet

Both models share identical reels, hit frequency and trigger rate and differ only by a single win-magnitude scale factor — confirming the model switch changes payback without altering the game shape. The player-facing Help screen reads the active model at runtime (97.00% shown for Standard), so the displayed RTP can never drift from the math.

TEST / CHECK	RESULT	EVIDENCE & NOTES
RTP WITHIN TARGET (BOTH MODELS)	PASS	STANDARD 97.07% / LEAN 84.59% OVER 10M SPINS EACH.
Paytable values match the par sheet	PASS	HELP paytable and dev MATH panel both read SYMBOLS[].pays; spot-checked end-to-end.
Bonus-buy price is fair	PASS	Configured 88× vs simulated fair 88.12×.
Displayed RTP matches configuration	PASS	HELP shows "RTP): 97.00%" for the active Standard model.

4. Core Gameplay

TEST / CHECK	RESULT	EVIDENCE & NOTES
INITIAL STATE ON LOAD	PASS	BALANCE \$1,000.00, BET \$1.00, WIN \$0.00; 15 SYMBOL CELLS (5×3) RENDERED.
Single spin deducts the bet	PASS	Balance \$1,000.00 → \$999.00 on a \$1.00 spin (exact).
Spin settles and re-enables controls	PASS	Spin button re-enabled after the reels stop.
Bet increase clamps at maximum	PASS	Repeated increase stops at \$50.00.
Bet decrease clamps at minimum	PASS	Repeated decrease stops at \$0.20; increments step correctly (0.20 → 0.50 → 1.00 ...).
Win evaluation (243 ways)	PASS	Wins credited to balance; verified via the bonus round payout below and the 10M-spin hit rate.
Idle status ticker	PASS	Rotating marketing lines display, including the config-derived "123X JACKPOT" line.

5. Bonus Feature — Hard-Hat Free Spins

The bonus was driven end-to-end through the Buy Bonus path (the most direct way to exercise the full feature deterministically).

TEST / CHECK	RESULT	EVIDENCE & NOTES
BUY BONUS COST IS CORRECT	PASS	CONFIRMATION SHOWS \$88.00 FOR A \$1.00 BET (88× — MATCHES BONUS_CONFIG).
Buy Bonus deducts the cost	PASS	Balance \$999.00 → \$911.00 (exactly −\$88.00) on confirm.
Bonus enters and runs free spins	PASS	"6 FREE GAMES REMAINING" counter, house construction (e.g. "STRAW HOUSE → \$1.73").
Houses build and award	PASS	Straw/stick/brick frames award per the bonus par sheet during the round.
Bonus completes and credits the win	PASS	"BONUS COMPLETE — WON \$94.00!"; balance \$911.00 + \$94.00 = \$1,005.00 (exact).
Controls restored after bonus	PASS	Spin button re-enabled; base game resumes cleanly.
Buy Bonus responsible-play note	PASS	Modal states the buy carries the same ~97% long-run payback as normal play.

6. Audio

TEST / CHECK	RESULT	EVIDENCE & NOTES
SOUND-EFFECT COVERAGE	PASS	EVERY SYNTH METHOD MAPS TO AN EXISTING FILE IN ASSETS/AUDIO/SFX/ (REELS, WINS, COINS, WOLF, WIND, FRAMES, UI).
Music tracks present & wired	PASS	bgm_base, bgm_bonus (crossfaded base↔bonus) and showdown_theme (High-Noon outro).
Narrator voice coverage	PASS	66 phrase categories, 565 lines; each category has a generated clip set.
Volume controls (SFX / Music / Voice)	PASS	Three independent sliders update audio and the % readouts live.

TEST / CHECK	RESULT	EVIDENCE & NOTES
MUTE ALL is a toggle	PASS	Stores levels, zeroes all three, relabels to UNMUTE, and restores on second press.
RESET restores defaults	PASS	Returns to 100 / 50 / 80 and clears mute state.
No overlapping voice lines	PASS	Narrator queue is single-track with cooldowns; High-Noon mutes ambient VO so only the scripted line plays.

7. User Interface & Layout

TEST / CHECK	RESULT	EVIDENCE & NOTES
GAME IS VERTICALLY CENTRED	PASS	AT 932×430 THE CABINET SITS 6PX TOP / 6PX BOTTOM — EQUAL MARGINS, NO SCROLL.
Public drawer is compact	PASS	Six buttons (Auto, Turbo, Volume, Buy Bonus, Help, Rules) at a uniform 34px height.
Developer drawer fits the screen	PASS	17 tools laid out in two columns at a uniform 30px; no squishing, no scroll, no overflow.
Drawer buttons uniform height	PASS	Plain buttons and the Volume/Time wrappers share a CSS height variable.
Fly-out panels stay on-screen	PASS	Volume/Time pop-outs open to the left of the drawer and are fully within the viewport in both builds.
No element overlap	PASS	Side wolf, borders, SPIN button, drawer and reels checked at the target landscape size.
Portrait orientation gate	PASS	"Rotate your device" guidance shown in portrait; game intended for landscape.

8. Help / Rules / Paytable

TEST / CHECK	RESULT	EVIDENCE & NOTES
HELP MODAL OPENS WITH TABS	PASS	HOW WINS PAY, PLUS A PAGED INTERFACE.
Rules modal opens with tabs	PASS	How Wins Pay · Paytable · Wolf Wild · Hard Hats · Bonus Feature · Rules.
Symbol names match the art	PASS	Paytable shows Shot Glass, Sheriff Badge, Horseshoe, Tornado (no legacy "Wood Pig").
243-ways explanation present	PASS	Win rule, adjacency and "ways = product across reels" documented.
RTP statement present & correct	PASS	Displays the active model RTP (97.00%).
No invented odds / no cert claims	PASS	All numbers traced to config; nothing claims certification.

9. Interactive Content & Easter Eggs

TEST / CHECK	RESULT	EVIDENCE & NOTES
SALOON SIGN CLICK	PASS	FIRES THE "SALOONHEADER" VOICE LINE (PRIORITY 58).
Vegas Low Roller medallion click	PASS	First click forces the signature line (vlrMedallion index 0).
Sheriff badge click	PASS	First click forces the signature line (sheriffBadge index 0).
Reel symbol click jokes	PASS	Clicking a symbol triggers its object-specific joke pool (suppressed during bonus).
High-Noon standoff sequence	PASS	Card + wolf voiceover → standoff clip → showdown theme over the card → gunshot cut-to-black → gameplay; contained in the reel window with wolf & borders visible.
Spin-hover taunt	PASS	Wired to fire when the SPIN button is hovered >3s without clicking (desktop).

10. Developer Tools

Developer tools are hidden on the public build and revealed with ?dev=1 (remembered in local storage).

TEST / CHECK	RESULT	EVIDENCE & NOTES
RTP PICKER (MATH MODEL)	PASS	RTP-MODAL OPENS.
Game Size breakdown	PASS	size-modal opens.
Add Credit (deposit)	PASS	deposit-modal opens and credits the balance.
Web Push (deploy link)	PASS	webpush-modal opens.
Monte-Carlo Simulator	PASS	sim-modal opens and renders results.
Math breakdown	PASS	math-modal opens; shows correct symbol labels (Shot Glass).
SEO info	PASS	seo-modal opens.
Time-of-day panel	PASS	time-panel opens.
Force Extreme Anticipation	PASS	Fires without error.
Force Expanding Wild	PASS	Fires without error.
Show Animation names	PASS	Fires without error.
Strike High Noon	PASS	Triggers the standoff sequence (see Section 9).

11. Compatibility, Performance, Accessibility, SEO & Compliance

iOS / Safari compatibility

TEST / CHECK	RESULT	EVIDENCE & NOTES
TRANSPARENT CLIPS HAVE HEVC-ALPHA TWINS	PASS	EVERY TRANSPARENT VIDEO (SIDE WOLF SET, SPIN, HOUSE FRAMES) SHIPS BOTH .WEBM AND A HEVC-ALPHA .MP4; WEBKIT IS SERVED THE .MP4 VIA VIDEO-FORMAT.JS.

TEST / CHECK	RESULT	EVIDENCE & NOTES
Poster frames for paused video	PASS	spin_poster.webp and wolf_poster.webp provided (iOS shows posters when a video is not actively painting).
Opaque cut-scene clips	INFO	Intro/standoff/tornado ship as .webm only (no alpha needed). Recommend confirming playback on a physical iOS device — see Section 12.

Performance / load

FIRST-PLAY DOWNLOAD	≈16.7 MB ACROSS 68 FILES (LOADS BEFORE THE FIRST SPIN)
Deferred (progressive) media	≈134.3 MB across 639 files (loaded lazily / on demand)
Developer-only assets	≈0.9 MB across 60 files (never sent to players)
Total project footprint	≈151.9 MB, 767 files

Accessibility & SEO

TEST / CHECK	RESULT	EVIDENCE & NOTES
CRAWLABLE TEXT FOR A JS GAME	PASS	SCREEN-READER HEADER (.SR-ONLY) WITH H1, DESCRIPTION AND FEATURE LIST.
ARIA labelling	PASS	26 aria-labels across controls; reel images carry alt text.
Head metadata	PASS	Title, description, canonical, theme-color, manifest, Open Graph and Twitter cards all present.
Structured data	PASS	Schema.org VideoGame JSON-LD.
Crawl directives	PASS	robots.txt and sitemap.xml present; 1200×630 share image (og-image.png).

Responsible-play compliance

TEST / CHECK	RESULT	EVIDENCE & NOTES
FRAMED AS FREE / VIRTUAL	PASS	"PLAYS FREE WITH VIRTUAL CREDITS — FUN CASINO-STYLE GAMEPLAY WITH NO REAL-MONEY WAGERING."

TEST / CHECK	RESULT	EVIDENCE & NOTES
No misleading or pressuring language	PASS	No "you're due", no guaranteed-win or "can't lose" wording; cold-streak lines are gentle by design.
No certification claims	PASS	Help explicitly avoids implying certification.

12. Defects, Observations & Recommendations

No blocking, major or minor functional defects were found in this pass. The following are advisory.

Observations (low severity)

- **Physical iOS Safari validation (Advisory).** Browser tests ran in a headless Chromium viewport emulating an iPhone in landscape, not on real WebKit hardware. Recommend a short pass on an actual iPhone (Safari) to confirm: video alpha/transparency, that the opaque .webm cut-scenes (intro, High-Noon standoff) play, audio autoplay-after-gesture, and the two-column developer drawer.
- **Add Credit increment (Advisory, developer-only).** The ADD CREDIT dev tool credited a small amount in automated testing; worth a human confirming the intended deposit amount. This affects only the hidden developer build, never players.
- **Bonus round duration (By design).** The free-spins feature is a rich, multi-second animated sequence. This is expected presentation, not a hang; no errors occur during it.

Recommendations for hand-off

- Run the headless math verifier (node tools/sim.js 50000000) before any release as the canonical RTP sign-off.
- Keep all player-facing numbers config-derived (as the payable, RTP and jackpot ticker now are) so they cannot drift from the math.
- If a custom domain is added, update the canonical URL, og:url, robots.txt and sitemap.xml to match.
- Verify the site in Google Search Console and submit the sitemap.

13. Sign-off

Based on the tests above, Big Bad Wolf Saloon is in a stable, internally-consistent and release-ready state for continued client testing. Mathematics match their targets, all features function, and no runtime errors were observed. The only outstanding item is confirmatory testing on a physical iOS Safari device, which is recommended before a public launch.

QA verdict: PASS · Defects: 0 blocking / 0 major / 0 minor · Advisories: 2